



BSc IN AERONAUTICAL ENGINEERING (AVIONICS) COURSE CURRICULUM

Structure of the Curriculum:

a. Duration of the Program: Year: 4 (Semester: 08)

b. Admission Requirements:

Admission in Bachelor Program:

1. Four Year BSc Program: The admission to the various B.Sc. programs of the University will be done for each session as decided by the Academic Council.
2. Eligibility:
 - a. A candidate must pass SSC/Equivalent and HSC/Equivalent examination from science group. Candidates passed in the current year and previous one year eligible to apply.
 - b. Candidates from science group must obtain a minimum GPA of 4.50 each in SSC/Equivalent and HSC/Equivalent with minimum A Grade in Mathematics, Physics, Chemistry, and English/English language subjects.
 - c. Candidates from General Certificate of Education (GCE) in 'O' level examination background must pass in minimum 5 subjects. Candidates having 'C' or lower grades in more than 2 subjects out of 5 subjects including Mathematics, Physics, Chemistry, and English will not be eligible for admission test.
 - d. Candidates from General Certificate of Education (GCE) in 'A' level examination background must pass in minimum 3 subjects. Candidates having 'C' or lower grades in more than 1 subject out of 3 subjects including Mathematics, Physics and Chemistry will not be eligible for admission test.
 - e. Candidates for GCE (O & A level) background must have minimum 27 points in total based on the following scale:

Grade	Point
A*/A	5.00
B	4.00
C	3.50
D	3.00

- f) Candidates from International Baccalaureate (IB) must pass in minimum 6 (six) subjects having minimum 30 points based on the rating scale used in their curriculum (7, 6, 5, 4). Rating of 1, 2 and 3 will not be considered in point calculation.
- g) Students must have Mathematics in HSC as their compulsory/optional (4th subject) subject.
- h) Non-Resident Bangladeshi applicants from abroad shall apply through Embassies/High Commission. Educational qualifications are same as applicable for Bangladeshi students.

3. Bangabandhu Sheikh Mujibur Rahman Aviation and Aerospace University (BSMRAAU) will invite application from the eligible candidates for admission in Bachelor Program once in a year aiming to start the session from January or as decided by the Academic Council.
4. BSMRAAU will offer four years BSc Program with 30 seats or as decided by Admission Committee in each session. The selection of the students for Bachelor Programs will be on a competitive basis of past academic records, and performances in the admission test with Viva-voce (if required).
5. Students must show the original Transcripts/Mark Sheets at the time of admission.
6. Students who all are finally selected for admission have to undergo a medical examination to be declared fit for the respective courses.
7. The university authority reserves the right to cancel/modify/change any information given in this advertisement. For any query, applicants may communicate at contact@bsmraau.edu.bd.

Qualifying Criteria for International Students:

1. Eligibility:
 - a. Passed Higher Secondary School or Equivalent for admission into Under Graduate Programs and passed Under Graduate Program for admission into Post Graduate Program.
 - b. GPA requirement for both the programs minimum 4.50 in 5 scale grading system and minimum 3.50 in 4 scale grading system.
 - c. For Engineering Program applicant must have passed Higher Secondary School/Equivalent with Mathematics, Physics and Chemistry.
 - d. Provide statement of purpose (SOP) for admission.
 - e. Applicant will need to submit a list of two qualified referees with one in relation to academic attainment.
 - f. Provide supporting paper (certificate and transcript) in support of academic qualifications.
2. English Language Requirement: Provide proof of English Proficiency, if English is not the first language to the country of citizen, as under:
 - a. Minimum Score 6.5 in IELTS with No Score below 6.0 in individual part
 - b. Minimum Score 1650 in SAT out of 2400 and Minimum Score of 1110 in SAT out of 1600.
 - c. Minimum Score 86 in TOEFL with No Score below 21 in each skill band
 - d. Validity of English Language Proficiency will be 2 years from test date.
 - e. No admission test will be required for the students fulfilling the above-mentioned criteria. However, the selected candidate will be called for Viva-voce before granting the Confirmation of Enrolment (CoE).

Note: Eligibility and other requirements may be reviewed as and when required and are to be approved by the Academic Council.

c. Credit Requirement to complete the program Total Minimum: 154 Credits

d. Total Class Week in a Semester: 14 Weeks

e. Minimum CGPA Requirements for Graduation: 2.25

f. Maximum Academic Year of Completion: 6 Years

g. Category of Courses:

(I) List of Core Engineering Courses:

A total of 40 courses on avionics engineering topics are included in the program which covers 90 credit hours.

Course Code	Course Title	Type of Course	Contact Hours	Credit Hours	Semester
AVE 4101	Electrical Circuits Analysis I	Theory	3.00	3.00	1
AVE 4102	Electrical Circuits Analysis I Sessional	Sessional	3.00	1.50	1
AVE 4201	Electrical Circuits Analysis II	Theory	3.00	3.00	2
AVE 4202	Electrical Circuits Analysis II Sessional	Sessional	1.50	0.75	2
AVE 4301	Electronic Circuits I	Theory	3.00	3.00	3
AVE 4302	Electronic Circuits I Sessional	Sessional	3.00	1.50	3
AVE 4401	Electronic Circuits II	Theory	3.00	3.00	4
AVE 4402	Electronic Circuits II Sessional	Sessional	3.00	1.50	4
AVE 4403	Electric Machine and Drives	Theory	3.00	3.00	4
AVE 4404	Electric Machines and Drives Sessional	Sessional	3.00	1.50	4
AVE 4405	Digital Logic and Computer Design	Theory	3.00	3.00	4
AVE 4406	Digital Logic and Computer Design Sessional	Sessional	3.00	1.50	4
AVE 4501	Signals and Systems	Theory	3.00	3.00	5
AVE 4502	Signals and Systems Sessional	Sessional	1.50	0.75	5
AVE 4503	Microprocessor and Microcontroller Systems	Theory	3.00	3.00	5
AVE 4504	Microprocessor and Microcontroller Systems Sessional	Sessional	1.50	0.75	5
AVE 4505	Electromagnetic Field Theory	Theory	3.00	3.00	5
AVE 4510	Modelling & Simulation Sessional	Sessional	3.00	1.50	5
AVE 4600	Industrial Training	Sessional	(4 weeks)	1.00	6
AVE 4601	Communication Systems	Theory	3.00	3.00	6
AVE 4602	Communication Systems Sessional	Sessional	3.00	1.50	6
AVE 4603	Real Time Embedded Systems	Theory	3.00	3.00	6
AVE 4604	Real Time Embedded Systems Sessional	Sessional	1.50	0.75	6
AVE 4605	Microwave and Antennas	Theory	3.00	3.00	6
AVE 4607	Numerical Methods	Theory	3.00	3.00	6
AVE 4608	Numerical Methods Sessional	Sessional	1.50	0.75	6
AVE 4609	Control Systems	Theory	3.00	3.00	6
AVE 4610	Control Systems Sessional	Sessional	1.50	0.75	6
AVE 4701	Digital Signal Processing	Theory	3.00	3.00	7
AVE 4702	Digital Signal Processing Sessional	Sessional	1.50	0.75	7
AVE 4703	Aircraft Electrical, Instrument and Autopilot Systems	Theory	3.00	3.00	7
AVE 4704	Aircraft Electrical, Instrument and Autopilot Systems Sessional	Sessional	3.00	1.50	7
AVE 4705	Avionics System Design	Theory	3.00	3.00	7
AVE 4706	Avionics System Design Sessional	Sessional	1.50	0.75	7
AVE 4707	Engineering Ethics & Professionalism	Theory	2.00	2.00	7
AVE 4801	Radar System Engineering	Theory	3.00	3.00	8
AVE 4802	Radar System Engineering Sessional	Sessional	3.00	1.50	8
AVE 4803	Aircraft Communication, Navigation &	Theory	3.00	3.00	8

Course Code	Course Title	Type of Course	Contact Hours	Credit Hours	Semester
	Surveillance System				
AVE 4804	Aircraft Communication, Navigation & Surveillance System Sessional	Sessional	3.00	1.50	8
AVE 4700	Final Year Research and Design Project (FYRDP)	Sessional	9.00	4.50	7+8

(II) List of Other Engineering Courses:

A total of 13 courses (7 theory and 6 sessional) are included in the programme covering 30.25 credit hours.

Course Code	Course Title	Type of Course	Contact Hours	Credit Hours	Semester
ASE 4101	Introduction to Aeronautical Engineering	Theory	3.00	3.00	1
ASE 4102	Aeronautical Engineering Drawing	Sessional	3.00	1.50	1
CSE 4201	Computer Programming and Application	Theory	3.00	3.00	2
CSE 4202	Computer Programming and Application Sessional	Sessional	3.00	1.50	2
ASE 4202	Workshop Technology Sessional	Sessional	3.00	1.50	2
ASE 4341	Thermodynamics	Theory	3.00	3.00	3
ASE 4342	Thermodynamics Sessional	Sessional	1.50	0.75	3
ASE 4353	Engineering Mechanics (Statics & Dynamics)	Theory	4.00	4.00	3
CSE 4305	Data Structures and Algorithm	Theory	3.00	3.00	3
CSE 4306	Data Structures and Algorithm Sessional	Sessional	3.00	1.50	3
ASE 4513	Aerodynamics	Theory	3.00	3.00	5
ASE 4514	Aerodynamics Sessional	Sessional	3.00	1.50	5
ASE 4807	Industrial & Business Management	Theory	3.00	3.00	8

(III) List of Basic Sciences and Mathematics Courses:

A total of 5 mathematics courses and 2 basic sciences courses including 2 sessional are included in the program covering 23.25 credit hours.

Course Code	Course Title	Type of Course	Contact hours	Credit Hours	Semester
MAT 4101	Differential and Integral Calculus	Theory	3.00	3.00	1
MAT 4203	Ordinary and Partial Differential Equations	Theory	3.00	3.00	2
MAT 4305	Linear Algebra, Coordinate Geometry and Complex Variables	Theory	3.00	3.00	3
MAT 4407	Fourier and Laplace Transform	Theory	3.00	3.00	4
MAT 4509	Probability & Statistics	Theory	3.00	3.00	5
PHY 4101	Physics	Theory	3.00	3.00	1
PHY 4102	Physics Sessional	Sessional	3.00	1.50	1
CHM 4201	Chemistry	Theory	3.00	3.00	2
CHM 4202	Chemistry Sessional	Sessional	1.50	0.75	2

(IV) List of Humanities and Social Sciences (HASS) Courses:

A total of 4 courses from HASS are included in the programme covering 9.75 credit hours.

Course Code	Course Title	Type of Course	Contact hours	Credit Hours	Semester
HUM 4101	Communicative English	Theory	3.00	3.00	1
HUM 4102	Communicative English Sessional (Technical Report Writing)	Sessional	1.50	0.75	1
HUM 4203	Bangladesh Studies and Sociology	Theory	3.00	3.00	2
HUM 4413	Engineering Economics	Theory	3.00	3.00	4

(V) Elective Courses

Students of B.Sc. in Avionics Engineering can choose 2 elective courses along with 1 sessional covering 6.75 credit hours. Students can take optional courses from any of the following courses offered. Availability of elective courses is subject to availability of faculty and related laboratory facilities.

a) List of Departmental Electives Courses

Course No	Course Title	Type of course	Contact hours	Credit Hours
AVE 4721	Avionics Software Design	Theory	3.00	3.00
AVE 4723	Autonomous Systems, Guidance & Control	Theory	3.00	3.00
AVE 4725	Robotics and Mechatronics	Theory	3.00	3.00
AVE 4729	Remote Sensing and Digital Image Processing	Theory	3.00	3.00
AVE 4731	Avionics Human Factors Engineering	Theory	3.00	3.00
AVE 4733	RF Integrated Circuit	Theory	3.00	3.00
AVE 4821	Introduction to Satellite Communication Engineering	Theory	3.00	3.00
AVE 4822	Introduction to Satellite Communication Engineering Sessional	Sessional	1.50	0.75
AVE 4823	Space Avionics	Theory	3.00	3.00
AVE 4824	Space Avionics Sessional	Sessional	1.50	0.75
AVE 4825	VLSI Design	Theory	3.00	3.00
AVE 4826	VLSI Design Sessional	Sessional	1.50	0.75
AVE 4827	Optical Fibre Communication System	Theory	3.00	3.00
AVE 4828	Optical Fibre communication system Sessional	Sessional	1.50	0.75
AVE 4829	Avionics Cyber Security	Theory	3.00	3.00
AVE 4830	Avionics Cyber Security Sessional	Sessional	1.50	0.75
AVE 4831	Wireless Communication	Theory	3.00	3.00
AVE 4832	Wireless Communication Sessional	Sessional	1.50	0.75

b) List of Non-Departmental Elective Courses

Undergraduate students from the Department of Avionics Engineering can choose courses from the following list of Other/Aerospace engineering courses.

Course Code	Course Title	Semester	Contact Hours	Credit Hours
ASE 4551	Aircraft Systems	Theory	3.00	3.00
ASE 4552	Aircraft Systems Sessional	Sessional	1.50	0.75

ASE 4557	Aerospace Vehicle Dynamics & Control	Theory	3.00	3.00
ASE 4797	Orbital Mechanics	Theory	3.00	3.00
CSE 4781	Machine learning	Theory	3.00	3.00
CSE 4783	Artificial Intelligence	Theory	3.00	3.00
CSE 4785	Introduction to Quantum Computing	Theory	3.00	3.00
EEE 4881	Electrical Power Systems	Theory	3.00	3.00
EEE 4882	Electrical Power Systems Sessional	Sessional	1.50	0.75

(VI) Final Year Research and Design Project (FYRDP)

Final Year Research and Design Project (FYRDP) will have to be undertaken by students under a supervisor in partial fulfilment of the requirement of his degree. Credit hours allotted to the Final Year Research and Design Project (FYRDP) will be 9 credit hours having 18 contact hours.

18. Semester Wise Distribution of Courses:

The detailed course plan for Bachelor of Science in Avionics Engineering is presented below. The programme includes **39 theory courses** and **31 sessional courses** covering **117** and **43 credit** hours for theory and sessional respectively. **Total credit hours** for the programme are **160** in 8 semesters.

Undergraduate students of the Department of Avionics Engineering (AVE) must undertake the following course schedule, the semester-wise distribution of which is given below:

First Year, First Semester

Course Code	Course Title	Type	Contact Hours	Credit Hours
PHY 4101	Physics	Theory	3.00	3.00
AVE 4101	Electrical Circuits Analysis I	Theory	3.00	3.00
MAT 4101	Differential and Integral Calculus	Theory	3.00	3.00
ASE 4101	Introduction to Aeronautical Engineering	Theory	3.00	3.00
HUM 4101	Communicative English	Theory	3.00	3.00
Subtotal (Theory)			15.00	15.00
PHY 4102	Physics Sessional	Sessional	3.00	1.50
AVE 4102	Electrical Circuits Analysis I Sessional	Sessional	3.00	1.50
ASE 4102	Aeronautical Engineering Drawing	Sessional	3.00	1.50
HUM 4102	Communicative English Sessional (Technical Report Writing)	Sessional	1.50	0.75
Subtotal (Sessional)			10.50	5.25
Total			25.50	20.25

First Year, Second Semester

Course Code	Course Title	Type	Contact Hours	Credit Hours
AVE 4201	Electrical Circuits Analysis II	Theory	3.00	3.00
CHM 4201	Chemistry	Theory	3.00	3.00
MAT 4203	Ordinary and Partial Differential Equations	Theory	3.00	3.00
CSE 4201	Computer Programming and Application	Theory	3.00	3.00
HUM 4203	Bangladesh Studies and Sociology	Theory	3.00	3.00
Subtotal (Theory)			15.00	15.00
AVE 4202	Electrical Circuits Analysis II Sessional	Sessional	1.50	0.75
CHM 4202	Chemistry Sessional	Sessional	1.50	0.75
ASE 4202	Workshop Technology Sessional	Sessional	3.00	1.50

Course Code	Course Title	Type	Contact Hours	Credit Hours
CSE 4202	Computer Programming and Application Sessional	Sessional	3.00	1.50
Subtotal (Sessional)			9.00	4.50
Total			24.00	19.50

Second Year, First Semester

Course Code	Course Title	Type	Contact hours	Credit Hours
AVE 4301	Electronic Circuit I	Theory	3.00	3.00
ASE 4341	Thermodynamics	Theory	3.00	3.00
ASE 4353	Engineering Mechanics (Statics & Dynamics)	Theory	4.00	4.00
MAT 4305	Linear Algebra, Coordinate Geometry and Complex Variables	Theory	3.00	3.00
CSE 4305	Data Structures and Algorithm	Theory	3.00	3.00
Subtotal (Theory)			16.00	16.00
AVE 4302	Electronic Circuit I Sessional	Sessional	3.00	1.50
ASE 4342	Thermodynamics Sessional	Sessional	1.50	0.75
CSE 4306	Data Structures and Algorithm Sessional	Sessional	3.00	1.50
Subtotal (Sessional)			7.50	3.75
Total			23.50	19.75

Second Year, Second Semester

Course Code	Course Title	Type	Contact Hours	Credit Hours
AVE 4401	Electronic Circuits II	Theory	3.00	3.00
AVE 4403	Electric Machines and Drives	Theory	3.00	3.00
MAT 4407	Fourier and Laplace Transform	Theory	3.00	3.00
AVE 4405	Digital Logic and Computer Design	Theory	3.00	3.00
HUM 4413	Engineering Economics	Theory	3.00	3.00
Subtotal (Theory)			15.00	15.00
AVE 4402	Electronic Circuits II Sessional	Sessional	3.00	1.50
AVE 4404	Electric Machines and Drives Sessional	Sessional	3.00	1.50
AVE 4406	Digital Logic and Computer Design Sessional	Sessional	3.00	1.50
Subtotal (Sessional)			9.00	4.50
Total			24.00	19.50

Third Year, First Semester

Course Code	Course Title	Type	Contact Hours	Credit Hours
AVE 4501	Signals and Systems	Theory	3.00	3.00
AVE 4503	Microprocessor and Microcontroller Systems	Theory	3.00	3.00
AVE 4505	Electromagnetic Field Theory	Theory	3.00	3.00
ASE 4513	Aerodynamics	Theory	3.00	3.00
MAT 4509	Probability & Statistics	Theory	3.00	3.00
Subtotal (Theory)			15.00	15.00
AVE 4502	Signals and Systems Sessional	Sessional	1.50	0.75
AVE 4504	Microprocessor and Microcontroller Systems Sessional	Sessional	1.50	0.75
AVE 4510	Modelling & Simulation Sessional	Sessional	3.00	1.50
ASE 4514	Aerodynamics Sessional	Sessional	3.00	1.50
Subtotal (Sessional)			9.00	4.50
Total			24.00	19.50

Third Year, Second Semester

Course Code	Course Title	Type	Contact hours	Credit Hours
AVE 4601	Communication Systems	Theory	3.00	3.00
AVE 4603	Real Time Embedded Systems	Theory	3.00	3.00
AVE 4605	Microwave and Antennas	Theory	3.00	3.00
AVE 4607	Numerical Methods	Theory	3.00	3.00
AVE 4609	Control Systems	Theory	3.00	3.00
Subtotal (Theory)			15.00	15.00
AVE 4600	Industrial Training	Sessional	(4 week)	1.00
AVE 4602	Communication Systems Sessional	Sessional	3.00	1.50
AVE 4604	Real Time Embedded Systems Sessional	Sessional	1.50	0.75
AVE 4608	Numerical Methods Sessional	Sessional	1.50	0.75
AVE 4610	Control Systems Sessional	Sessional	1.50	0.75
Subtotal (Sessional)			9.50	4.75
Total			24.50	19.75

Fourth Year, First Semester

Course Code	Course Title	Type	Contact hours	Credit Hours
AVE 4701	Digital Signal Processing	Theory	3.00	3.00
AVE 4703	Aircraft Electrical, Instrument and Autopilot Systems	Theory	3.00	3.00
AVE 4705	Avionics System Design	Theory	3.00	3.00
AVE 4707	Engineering Ethics & Professionalism	Theory	2.00	2.00
AVE 47XX	Elective - I	Theory	3.00	3.00
Subtotal (Theory)			14.00	14.00
AVE 4702	Digital Signal Processing Sessional	Sessional	1.50	0.75
AVE 4704	Aircraft Electrical, Instrument and Autopilot Systems Sessional	Sessional	3.00	1.50
AVE 4706	Avionics System Design Sessional	Sessional	1.50	0.75
AVE 4700	Final Year Research and Design Project (FYRDP)	Sessional	9.00	4.50
Subtotal (Sessional)			15.00	7.50
Total			29.00	21.50

Fourth Year, Second Semester

Course Code	Course Title	Type	Contact hours	Credit Hours
AVE 4801	Radar System Engineering	Theory	3.00	3.00
AVE 4803	Aircraft Communication, Navigation & Surveillance System	Theory	3.00	3.00
ASE 4807	Industrial & Business management	Theory	3.00	3.00
AVE 48XX	Elective - II	Theory	3.00	3.00
Subtotal (Theory)			12.00	12.00
AVE 4802	Radar System Engineering Sessional	Sessional	3.00	1.50
AVE 4804	Aircraft Communication, Navigation & Surveillance System Sessional	Sessional	3.00	1.50
AVE 48XX	Elective - II Sessional	Sessional	1.50	0.75
AVE 4700	Final Year Research and Design Project (FYRDP)	Sessional	9.00	4.50
Subtotal (Sessional)			16.50	8.25

Course Code	Course Title	Type	Contact hours	Credit Hours
Total			28.50	20.25

19. Summary of Departmental, Non-departmental, Basic Sciences, Mathematics and Humanities Courses

Semester	Theory		Sessional		Total	
	Credit Hours	Contact Hours	Credit Hours	Contact Hours	Credit Hours	Contact Hours
Semester 1	15.00	15.00	5.25	10.50	20.25	25.50
Semester 2	15.00	15.00	4.50	9.00	19.50	24.00
Semester 3	16.00	16.00	3.75	7.50	19.75	23.50
Semester 4	15.00	15.00	4.50	9.00	19.50	24.00
Semester 5	15.00	15.00	4.50	9.00	19.50	24.00
Semester 6	15.00	15.00	4.75	9.50	19.75	24.50
Semester 7	14.00	14.00	7.50	15.00	21.50	29.00
Semester 8	12.00	12.00	8.25	16.50	20.25	28.50
Total	117.00	117.00	43.00	86.00	160.00	203.00

20. Distribution of Credit Hours for Different Categories of Courses

Semester	Humanities	Math	Basic	Engineering		Elective	FYRDP	Total
				Core	Other			
Semester 1	3.75	3.00	4.50	4.50	4.50	-	-	20.25
Semester 2	3.00	3.00	3.75	3.75	6.00	-	-	19.50
Semester 3	-	3.00	-	4.50	12.25	-	-	19.75
Semester 4	3.00	3.00	-	13.50	-	-	-	19.50
Semester 5	-	3.00	-	12.00	4.50	-	-	19.50
Semester 6	-	-	-	19.75	-	-	-	19.75
Semester 7	-	-	-	14.00	-	3.00	4.50	21.50
Semester 8	-	-	-	9.00	3.00	3.75	4.50	20.25
Total Credit Hours	9.75	15.00	8.25	81.0	30.25	6.75	9.00	160.00
	6.09%	9.37%	5.16%	50.63%	18.9%	4.22%	5.63%	100%

21. List of Elective Courses

a. Elective -I [One Theory]

Course No	Course Title	Type	Contact hours	Credit Hours
AVE 4721	Avionics Software Design	Theory	3.00	3.00
AVE 4723	Autonomous Systems, Guidance & Control	Theory	3.00	3.00
AVE 4725	Robotics and Mechatronics	Theory	3.00	3.00
AVE 4729	Remote Sensing and Digital Image Processing	Theory	3.00	3.00
AVE 4731	Avionics Human Factors Engineering	Theory	3.00	3.00
AVE 4733	RF Integrated Circuit	Theory	3.00	3.00
CSE 4781	Machine Learning	Theory	3.00	3.00
CSE 4783	Artificial Intelligence	Theory	3.00	3.00
CSE 4785	Introduction to Quantum Computing	Theory	3.00	3.00

ASE 4557	Aerospace Vehicle Dynamics & Control	Theory	3.00	3.00
ASE 4797	Orbital Mechanics	Theory	3.00	3.00

b. Elective-II [One Theory + One Sessional]

Course No	Course Title	Type	Contact hours	Credit Hours
AVE 4821	Introduction to Satellite Communication Engineering	Theory	3.00	3.00
AVE 4823	Space Avionics	Theory	3.00	3.00
AVE 4825	VLSI Design	Theory	3.00	3.00
AVE 4827	Optical Fibre Communication System	Theory	3.00	3.00
AVE 4829	Avionics Cyber Security	Theory	3.00	3.00
AVE 4831	Wireless Communication	Theory	3.00	3.00
ASE 4551	Aircraft Systems	Theory	3.00	3.00
AVE 4822	Introduction to Satellite Communication Engineering Sessional	Sessional	1.50	0.75
AVE 4824	Space Avionics Sessional	Sessional	1.50	0.75
AVE 4826	VLSI Design Sessional	Sessional	1.50	0.75
AVE 4828	Optical Fibre communication system Sessional	Sessional	1.50	0.75
AVE 4830	Avionics Cyber Security Sessional	Sessional	1.50	0.75
AVE 4832	Wireless Communication Sessional	Sessional	1.50	0.75
ASE 4552	Aircraft Systems Sessional	Sessional	1.50	0.75